

Assignment 6

Due: 3/18

Note: Show all your work.

Problem 1 (10 points). Using the *glm* model in R, build a logistic model from the *hw6_p1.csv* dataset.

- (1). Show all coefficients of the model for Class 1.
- (2). Suppose you are classifying a new object $O = \langle \text{ClosePrice} = 30, \text{OpenPrice} = 15.00 \rangle$ yourself (not using R). Calculate the probability of the new object belonging to Class 1 and classify it assuming the classification threshold is 0.5. You must do calculation yourself (i.e., do not use R or any other software that calculates probabilities for you).

The following three problems are practice of parameter tuning in R.

Problem 2 (10 points). Use *hw_p2.csv* dataset for this problem. This dataset was downloaded from the UCI ML Repository and modified for this assignment.

Build and test a J48 model according to the following requirements:

- Split the dataset into *train* and *test* sets with the 66%-34% ratio.
- Perform 10-fold cross-validations 5 times (to find the best parameter values).
- Use *trainControl*, *tuneGrid*, and *train* functions.
- Find the best parameter values (J48 has two parameters – C and M).
- Test the model on the test dataset.

In your submission file, you must include the following:

- Best values of C and M
- Confusion matrix resulting from the test on the test dataset.

You must also submit your R code file. Make sure to set seed, whenever necessary, with *set.seed(31)* so that the TA can reproduce your result.

Problem 3 (10 points). Use *hw6_p2.csv* dataset for this problem (the same dataset used for Problem 2). This dataset was downloaded from the UCI ML Repository and modified for this assignment.

Build and test a *rpart* decision tree model according to the following requirements:

- Split the dataset into *train* and *test* sets with the 66%-34% ratio.
- Perform 10-fold cross-validations 5 times (to find the best parameter value).
- Use *trainControl*, *tuneLength*, and *train* functions.

- Find the best parameter value (*rpart* has one parameter – *cp*).
- Test the model on the test dataset.

In your submission file, you must include the following:

- Best value of *cp*
- Confusion matrix resulting from the test on the test dataset

You must also submit your R code file. Make sure to set seed, whenever necessary, with *set.seed(31)* so that the TA can reproduce your result.

Problem 4 (10 points). Use *hw6_p2.csv* dataset for this problem (the same dataset used for Problem 3). This dataset was downloaded from the UCI ML Repository and modified for this assignment.

Build and test a *KNN* model according to the following requirements:

- Split the dataset into *train* and *test* sets with the 66%-34% ratio.
- Perform 10-fold cross-validations 5 times (to find the best parameter values).
- Use *trainControl*, *tuneGrid*, and *train* functions.
- Find the best parameter value (*knn* has one parameter – *k*).
- Test the model on the test dataset.

In your submission file, you must include the following:

- Best value of *k*
- Confusion matrix resulting from the test on the test dataset

You must also submit your R code file. Make sure to set seed, whenever necessary, with *set.seed(31)* so that the TA can reproduce your result.

Submission:

Include all answers in a single file and name it *LastName_FirstName_HW6.EXT*. Here, “EXT” is an appropriate file extension (e.g., docx or pdf). If you have multiple files, then combine all files into a single archive file. Name the archive file as *LastName_FirstName_HW6.EXT*. Here, “EXT” is an appropriate archive file extension (e.g., zip or rar). Upload the file to Blackboard.